Shihan Kanungo

shihankanungo@gmail.com | | webpage | US citizen

EDUCATION

Henry M Gunn High School GPA 3.96/4.46W | **San Josè State University (SJSU)** GPA 4.0 . . . . . *2022–2026*

SELECTED HONORS

- 17th Romanian Master in Mathematics. Bronze medal ('26)
- USAMO (USA Math Olympiad) ('22 '25 '26); Gold award ('25 '26)
- MAA's Math Olympiad Program, 'Blue' MOP ('25)
- USAPhO (USA Physics Olympiad) Gold Medal & US Physics Team ('24)
- USAMTS (USA Math Talent Search) Gold medal ('23, '25)
- The William Lowell Putnam Competition. HM ('24) N2 ('25)
- S.-T. Yau Science Award. USA Bronze medal in Mathematics ('24)
- Spirit of Ramanujan $2k Fellowship ('23)
- WSS, World Science Scholar ('24)
- 2025 Global Child Prodigy Award. Recipient ('25)

RESEARCH HIGHLIGHTS

MIT PRIMES-USA (Combinatorics, Representation Theory, Commutative Algebra)

- Upper Bounds for Sequence Saturation [arXiv] Submitted to *Discrete Math.* (2025)
- Twisted Yangians in Complex Rank (with A.S. Kannan) [arXiv] Submitted to *Algebr. Represent. Th* (2025)
- Mixed Tensor Products ... and Newton's Formula for $\mathfrak{gl}(m|n)$ (with S. Erat, A.S. Kannan) [arXiv] *Transform. Groups.* (2025)
- The Length Finite Factorization Property. (with H. Jiang, H. Kim) [arXiv] *Commun. Korean Math. Soc.* **39** (2024), no. 2, 313-329. (2024)

San José State University (SJSU) (Number Theory)

- On Product Formulas of Guillera and Sondow (with J. Schettler) [arXiv] Submitted to *The Ramanujan Journal* (2024)

ADVANCED COURSEWORK

San José State University (Graduate courses shown in color)

A	MATH 221A *Higher Algebra I*	Edgar Bering	*'25 F*
A+	MATH 226 Theory of Numbers	Jordan Schettler	*'24 Sp*
A+	MATH 285M *Elliptic Curves & Modular Forms*	J Schettler	*'25 F*
A+	MATH 229 *Advanced Matrix Theory*	Wasin So	*'24 F*
A+	MATH 285M *Nonnegative Matrices*	Wasin So	*'25 Sp*
A+	MATH 275A *Topology*	Yan Zhang	*'25 F*
A+	MATH 279A *Graph Theory*	Yan Zhang	*'24 F*
A+	MATH 231A *Real Analysis*	Slobodan Simic	*'24 F*
A+	MATH 231B *Functional Analysis*	Slobodan Simic	*'25 Sp*
A+	MATH 233A *Applied Mathematics I*	Daniel Brinkmann	*'25 Sp*
A+	MATH 234 Advanced Dynamical Systems	Liam Stanton	*'24 Sp*
A	MATH 285A *Stochastic Methods*	Liam Stanton	*'25 Sp*
A	MATH 39 Linear Algebra I	Dashiell Fryer	*'23 F*
A+	MATH 129B Linear Algebra II	Wasin So	*'24 Sp*
A+	MATH 131A Introduction to Analysis	Tim Hsu	*'24 Sp*
A+	MATH 131B *Introduction to Real Variables*	Tim Hsu	*'24 F*
A+	MATH 138 Complex Variables	Liam Stanton	*'23 F*
A+	MATH 179 *Introduction to Graph Theory*	Sogol Jahanbekam	*'24 F*

Euler Circle (Grades not assigned)

Combinatorial Game Theory . . . . . *2024–'25*
Analytic Number Theory, Ergodic Theory, Number Theory . *2023–'24*
Differential Geometry, Generating Functions . . . . . *2021–'22*
Markov Chains, Abstract Algebra, Real Analysis . . . . . *2020–'21*

Art of Problem Solving (AoPS)

Group Theory (A+), Calculus (A+) *2020–23*

Technical Skills: Python, Java, MATLAB, LaTeX, Asymptote.

EXPOSITORY PAPERS

- *David Gale's Subset Takeaway Game,* Combinatorial Game Theory ('24)
- *Space-Filling Curves,* Real Analysis ('24)
- *The Hahn-Banach Theorem and Applications,* Functional Analysis ('25)
- *Sobolev Spaces and Applications to PDEs,* Theory of PDEs ('25)
- *The Stochastic Gradient Descent Method,* Stochastic Methods ('25)
- *Chromatic Polynomials,* Graph Theory ('24)
- *Lagrangian and Hamiltonian Mechanics,* Dynamical Systems ('24).
- *Large Gaps Between Primes,* Analytic Number Theory ('24)
- *Dynamical Billiards,* Ergodic Theory ('24)
- *Arithmetic Dynamics,* Number Theory ('23)
- *Minimal surfaces,* Differential Geometry ('22)
- *Herbert Wilf's snake-oil method,* Generating Functions ('21)
- *Mixing-time estimates for riffle shuffle,* Markov chains ('20)
- *Classifying groups of certain orders,* Algebra ('20)

CAMPS & PROGRAMS

- MOP (Math Olympiad Program), IMSA, Aurora IL. (Jun '25)
- USAPhO Team Training Camp, Univ. of Maryland. (Jun '24)
- The Special Lectures. Kharkiv-Vienna Intl. Sc. School. (Aug '23)

MUSIC

- Stanford Jazz Workshop (2024)
  Duke Ellington award; Outstanding Musicianship award
- Northern California Young Talents Competition (2023)
  First place, solo violin classical performance
- Golden State Youth Orchestra (GSYO)
  - Assistant concertmaster. Camilla Kolchinsky Orchestra (2023)
  - Concertmaster. Camerata & Chamber Players Orch. ('20–'22)

Languages: English (native) & French (beginner).